Let $U \subset \mathbb{R}^n$ be a bdd open set with $\partial U \in C^1$.

<u>Gagliardo-Nirenberg-Sobolev inequality:</u> If $1 \leq p < n$, then $W^{1,p}(U) \hookrightarrow L^{p^*}(U)$, where $p^* = \frac{np}{n-p}$, i.e.,

$$\|u\|_{L^{p^*}(U)} \leq C \|u\|_{W^{1,p}(U)} \quad \forall u \in W^{1,p}(U)$$

for some constant $C > 0$.

<u>Morrey's inequality:</u> If $n < p \leq \infty$, then $W^{1,p}(U) \hookrightarrow C^{0,\gamma}(\bar{U})$,

where $\gamma = 1 - \frac{n}{p}$, i.e.,

$$\|u\|_{C^{0,\gamma}(\bar{U})} \leq C \|u\|_{W^{1,p}(U)} \quad \forall u \in W^{1,p}(U)$$

for some constant $C > 0$.

$0 < \frac{n}{p} < 1$

## Embeddings for $W^{k,p}(U)$, $k \geq 1$, $1 \leq p \leq \infty$

**Thm (GNS inequality):** If $kp < n$, then $W^{1,p}(U) \hookrightarrow L^{p^*}(U)$,

where $p^* = \dfrac{np}{n-kp}$, i.e.,

$$\|u\|_{L^{p^*}(U)} \leq C \|u\|_{W^{k,p}(U)} \quad \forall u \in W^{k,p}(U)$$

$$\frac{1}{p^*} = \frac{n-kp}{np} = \frac{1}{p} - \frac{k}{n}$$

for some constant $C > 0$.

<u>Pf:</u> Let $u \in W^{k,p}(U)$. Then

$$\|u\|_{W^{k,p}(U)} = \left( \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(U)} \right)^{1/p} < \infty.$$

So for any $\beta$ with $|\beta| \leq k-1$, $\|D(D^\beta u)\|_{L^p(U)} < \infty$ and hence $D^\beta u \in W^{1,p}(U)$. By the GNS inequality for $W^{1,p}(U)$, then

$$\|D^\beta u\|_{L^{np/(n-p)}(U)} \leq C \|D^\beta u\|_{W^{1,p}(U)} \leq C \|u\|_{W^{k,p}(U)}$$

and hence $u \in W^{k-1, np/(n-p)}(U)$ and

$$\|u\|_{W^{k-1, np/(n-p)}(U)} \leq C \|u\|_{W^{k,p}(U)}.$$

Repeating this argument then shows that $u \in W^{k-2, np/(n-2p)}(U)$ and

$$\|u\|_{W^{k-2, np/(n-2p)}(U)} \leq C \|u\|_{W^{k-1, np/(n-p)}(U)}$$

$$\leq C \|u\|_{W^{k,p}(U)}.$$

$$\|u\|_{W^{k-1, np/(n-p)}(U)} = \left( \sum_{|\beta| \leq k-1} \|D^\beta u\|_{L^{np/(n-p)}(U)} \right)^{(n-p)/np}$$

$$\frac{n-p}{np} = \frac{1}{p} - \frac{1}{n}$$

$$\left(\frac{1}{p} - \frac{1}{n}\right) - \frac{1}{n} = \frac{1}{p} - \frac{2}{n} = \frac{n-2p}{np}$$

Repeating $k-2$ more times shows that $u \in L^{\overbrace{np/(n-kp)}^{p^*}}(U)$ $\Big|$ $\frac{1}{p} - \frac{k}{n} = \frac{n-kp}{np}$

and

$$\|u\|_{L^{p^*}(U)} \leq C \|u\|_{W^{1,p}(U)} \quad \#$$

<u>Ex:</u> $n=3,\ p=2,\ k=1,\ p^* = \frac{np}{n-p} = 6$

$$H^1(U) \hookrightarrow L^6(U)$$

**Thm (Morrey's inequality):** Let $kp > n$.

$0 < \frac{n}{p} < k$

Let $l = \left[\frac{n}{p}\right]$ be the integer part of $\frac{n}{p}$, i.e.,

$\frac{n}{p} = l + \gamma$

$l \leq \frac{n}{p} < l+1$, and let $\gamma = \frac{n}{p} - l \in [0,1)$.

1). If $\frac{n}{p}$ is not an integer, then $W^{k,p}(U) \hookrightarrow C^{k-l-1,1-\gamma}(\bar{U})$, i.e.,

$$\|u\|_{C^{k-l-1,1-\gamma}(\bar{U})} \leq C\|u\|_{W^{k,p}(U)} \quad \forall u \in W^{k,p}(U)$$

for some constant $C > 0$.

2). If $\frac{n}{p}$ is an integer, then $W^{k,p}(U) \hookrightarrow C^{k-\ell-1,\delta}(\bar{U})$

for any $\delta \in (0,1)$, i.e.,

$$\|u\|_{C^{k-\ell-1,\delta}(\bar{U})} \leq C\|u\|_{W^{k,p}(U)} \quad \forall u \in W^{k,p}(U)$$

for some constant $C > 0$.

Pf: 1). As in the proof of the GNS inequality,

$u \in W^{k-\ell,\, np/(n-\ell p)}(U)$ and

$$\|u\|_{W^{k-\ell,\, np/(n-\ell p)}(U)} \leq C \|u\|_{W^{k,p}(U)}$$

$$\frac{n-\ell p}{np} = \frac{1}{p} - \frac{\ell}{n}$$

$$\ell < \frac{n}{p} < \ell + 1$$

$$\frac{\ell}{n} < \frac{1}{p} < \frac{\ell}{n} + 1$$

$$0 < \frac{1}{p} - \frac{\ell}{n} < 1$$